

NANYANG TECHNOLOGICAL UNIVERSITYSEMESTER 1 EXAMINATION 2024-2025SC3103/CE3103 – EMBEDDED PROGRAMMING

Nov/Dec 2024

Time Allowed: 2 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 5 pages.
2. Answer **ALL** questions.
3. This is a closed-book examination.
4. Questions **DO NOT** carry equal marks.
5. **Appendix A** at the end of the exam paper contains schedulability tests that you may find useful while answering some of the questions in this exam.

1. Consider a real-time task set comprising tasks, $\tau_1, \tau_2, \dots, \tau_{10}$, where τ_1 has the highest priority and τ_{10} has the lowest priority. Suppose resources R_1, R_2, R_3 and R_4 , are shared by these tasks. The usage profile for these resources is as follows, where $RAT_{i,j}$ denotes maximum resource access time for resource R_j by task τ_i .

- $RAT_{2,1} = 3$;
- $RAT_{3,1} = 2$; $RAT_{3,4} = 3$;
- $RAT_{6,3} = 4$;
- $RAT_{8,3} = 6$; $RAT_{8,4} = 2$;
- $RAT_{9,1} = 1$; $RAT_{9,2} = 4$; $RAT_{9,4} = 3$;

Any $RAT_{i,j}$ not mentioned above by default has value 0 and it can be assumed that task τ_i does not use resource R_j .

- (a) Assuming Priority Ceiling Protocol (PCP) is used for resource management, compute the worst case blocking time for all tasks. Justify your answers.

(10 marks)

Note: Question 1 continues on page 2

- (b) Illustrate via a timing diagram how task τ_3 is blocked repeatedly while trying to access resource R_1 if Priority Inheritance Protocol (PIP) is used instead of PCP.

(10 marks)

2. Consider a periodic real-time task system comprising 6 tasks scheduled on a **dual-core** processor. Suppose the tasks in the system are specified as $\tau_1 = (10, 2, 10)$, $\tau_2 = (8, 2, 8)$, $\tau_3 = (7, 2, 7)$, $\tau_4 = (5, 2, 5)$, $\tau_5 = (4, 1, 4)$ and $\tau_6 = (2, 1, 2)$, where for a task $\tau_i = (T_i, C_i, D_i)$, T_i is the period, C_i is the worst-case execution time and D_i is the relative deadline.

- (a) Is this task system feasible on a dual-core processor? Justify your answer.

(2 marks)

- (b) Generate the schedule for this task system under **global EDF** scheduling algorithm for 10 time units. Can you conclude if this task system is schedulable under **global EDF**? Justify your answer.

(10 marks)

- (c) Suppose we wish to use partitioned scheduling strategies to schedule the above task system. While answering the below questions, show your workings in detail, including that of any schedulability tests employed.

- (i) Is there a partitioning strategy that would ensure schedulability under **EDF scheduling** on each core?

(8 marks)

- (ii) Is there a partitioning strategy that would ensure schedulability under **RM scheduling** on each core?

(10 marks)

3. (a) Explain every component in the following implicit rule of a Makefile:

```
% .o: %.c
    gcc -c $^ -o $@
```

(6 marks)

Note: Question 3 continues on page 3

(b) Explain the different types of processes in the Linux system. (6 marks)

(c) What is the command to run a process in the background? How do you terminate the process in the background? (3 marks)

(d) What is the expected output of the following code? Explain your answer and compare if sleep(1) is removed.

```
int g=10;
void* TFun(void* vargp)
{
    pthread_t tid1,tid2;
    int s = 0;
    ++s;

    if(g>0){
        --g;
        sleep(1);
        printf("s: %d, g: %d\n",s,g);
        pthread_create(&tid1,NULL,TFun,NULL);
        pthread_create(&tid2,NULL,TFun,NULL);

        pthread_join(tid1,NULL);
        pthread_join(tid2,NULL);
    }
}

int main()
{
    TFun(NULL);
    pthread_exit(NULL);
    return 0;
}
```

(10 marks)

4. (a) Compare and contrast the use of named pipes with other IPC mechanisms such as shared memory and message queues. In what scenarios would you prefer named pipes over these other methods? (9 marks)

- (b) Discuss the advantages of using Loadable Kernel Modules over a monolithic kernel design.
(4 marks)

- (c) In the Linux operating system, devices are classified into different types. Explain the differences between character devices and block devices. Provide an example of each type, and discuss how they interact with the Linux kernel.
(6 marks)

- (d) Explain the role of the bootloader in the boot process of a Raspberry Pi running an embedded Linux system.
(6 marks)

Appendix A (Schedulability Tests)

In the following equations,

- n denotes total number of tasks in the system,
- m denotes number of cores,
- $\tau_i = (T_i, C_i, T_i)$ denotes a periodic/sporadic task with period T_i , execution time C_i , and relative deadline T_i ,
- B_i denotes maximum blocking time for task τ_i ,
- $HP(\tau_i)$ is the set of all tasks τ_k such that $T_k \leq T_i$, **excluding task τ_i** and
- n_i denotes total number of tasks in $HP(\tau_i)$.

1. **Feasibility test:** $\sum_{i=1}^n C_i/T_i \leq m$
2. **Sufficient test under RM:** $\sum_{i=1}^n C_i/T_i \leq n(2^{1/n} - 1)$
3. **Exact test under RM:** For each task τ_i , $R_i \leq T_i$, where R_i is the response time of task τ_i given by the solution of the following recursive equation
 - $R_i(0) = C_i$, and
 - $R_i(x) = C_i + \sum_{\tau_k: \tau_k \in HP(\tau_i)} \left\lceil \frac{R_i(x-1)}{T_k} \right\rceil C_k$
4. **Exact test with blocking under EDF:** For each task τ_i , $\sum_{\tau_k: \tau_k \in HP(\tau_i)} C_k/T_k + C_i/T_i + B_i/T_i \leq 1$
5. **Sufficient test with blocking under RM:** For each task τ_i , $\sum_{\tau_k: \tau_k \in HP(\tau_i)} C_k/T_k + C_i/T_i + B_i/T_i \leq (n_i + 1)(2^{1/(n_i+1)} - 1)$
6. **Exact test with blocking under RM:** For each task τ_i , $R_i \leq T_i$, where R_i is given by the solution of the following recursive equation
 - $R_i(0) = C_i + B_i$, and
 - $R_i(x) = C_i + B_i + \sum_{\tau_k: \tau_k \in HP(\tau_i)} \left\lceil \frac{R_i(x-1)}{T_k} \right\rceil C_k$

END OF PAPER

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Please read the following instructions carefully:

- 1. Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**
2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.
3. Please write your Matriculation Number on the front of the answer book.
4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.